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VELLORE CAMPUS



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Task 4

MPMC LAB

Registration Number: 24BDS0320

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Experiment Number: 4

Date: 27 March 2026

AIM:

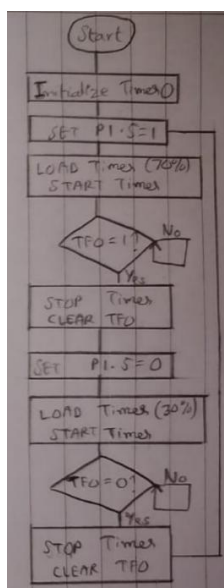
To write, compile and simulate an 8051 microcontroller program using Keil μ Vision software and verify the output using the simulator.

APPARTUS REQUIRED:

Keil μ Vision software

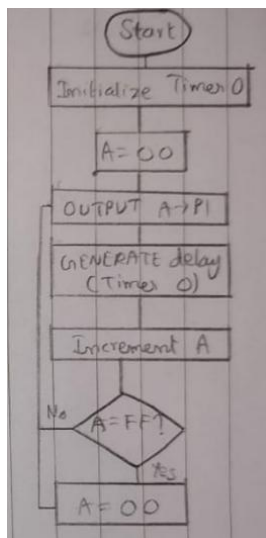
ALGORITHM:

1)



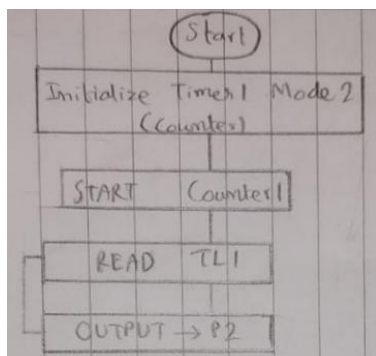
1. Start
2. Initialize Timer0 in Mode 1
3. Set P1.5 = 1 (HIGH)
4. Load Timer0 for ON delay (70%)
5. Start Timer0
6. Wait until TF0 = 1
7. Stop timer and clear TF0
8. Set P1.5 = 0 (LOW)
9. Load Timer0 for OFF delay (30%)
10. Start Timer0
11. Wait until TF0 = 1
12. Stop timer and clear TF0
13. Repeat from step 3

2)



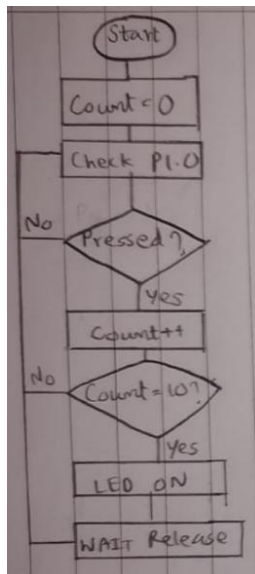
1. Start
2. Initialize register R0 = 00
3. Move R0 to Port P1
4. Generate delay using Timer0
5. Increment R0
6. If R0 $\neq$ FF $\rightarrow$ go to step 3
7. Else reset R0 = 00
8. Repeat

3)



1. Start
2. Initialize Timer1 in Mode 2 (Counter mode)
3. Load TH1 = 00
4. Start Counter1
5. Read TL1 value
6. Output TL1 to Port P2
7. Repeat continuously

4)



1. Start
2. Initialize count = 0
3. Check input switch (P1.0)
4. If switch pressed → increment count
5. Compare count with 10
6. If count = 10 → turn ON LED (P2.0)
7. Wait until switch released (debounce)
8. Repeat

CODE:

1)

```

ORG 0000H
MAIN:
SETB P1.5
ACALL DELAY70
CLR P1.5
ACALL DELAY30
SJMP MAIN
DELAY70:
MOV TMOD, #01H
MOV TH0, #0E6H
MOV TLO, #0A8H
SETB TRO
WAIT1:
JNB TFO, WAIT1
CLR TRO
CLR TFO
RET
DELAY30:
MOV TMOD, #01H
MOV TH0, #0F6H
MOV TLO, #0A8H
SETB TRO
WAIT2:
JNB TFO, WAIT2
CLR TRO
CLR TFO
RET
END
  
```

2)

```
ORG 0000H
MAIN:
MOV A, #00H
UP:
MOV P1, A
INC A
CJNE A, #0FFH, UP
SJMP MAIN
END
```

3)

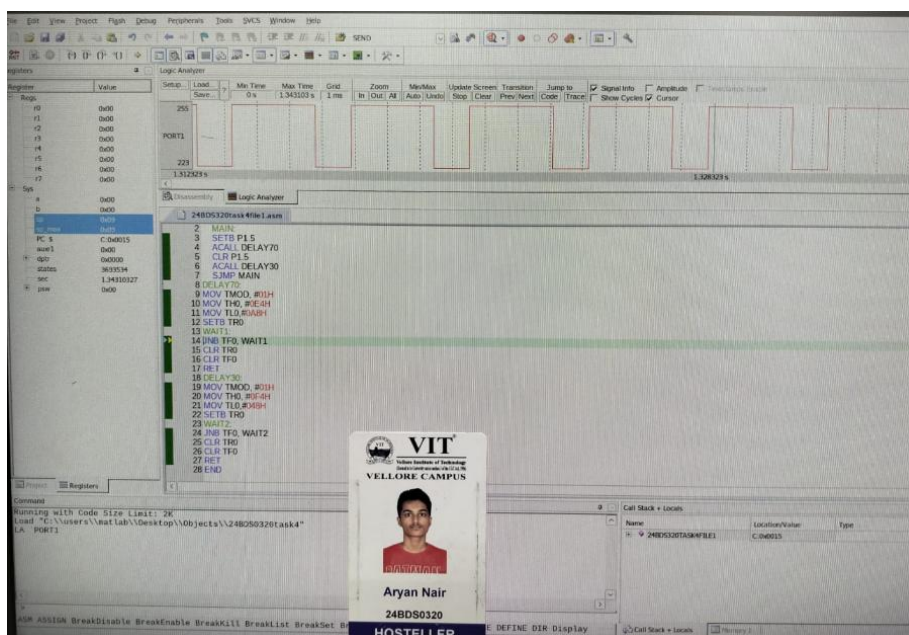
```
ORG 0000H
MOV TMOD, #60H
MOV TH1, #00H
MOV TL1, #00H
SETB TR1
MAIN:
MOV A, TL1
MOV P2, A
SJMP MAIN
END
```

4)

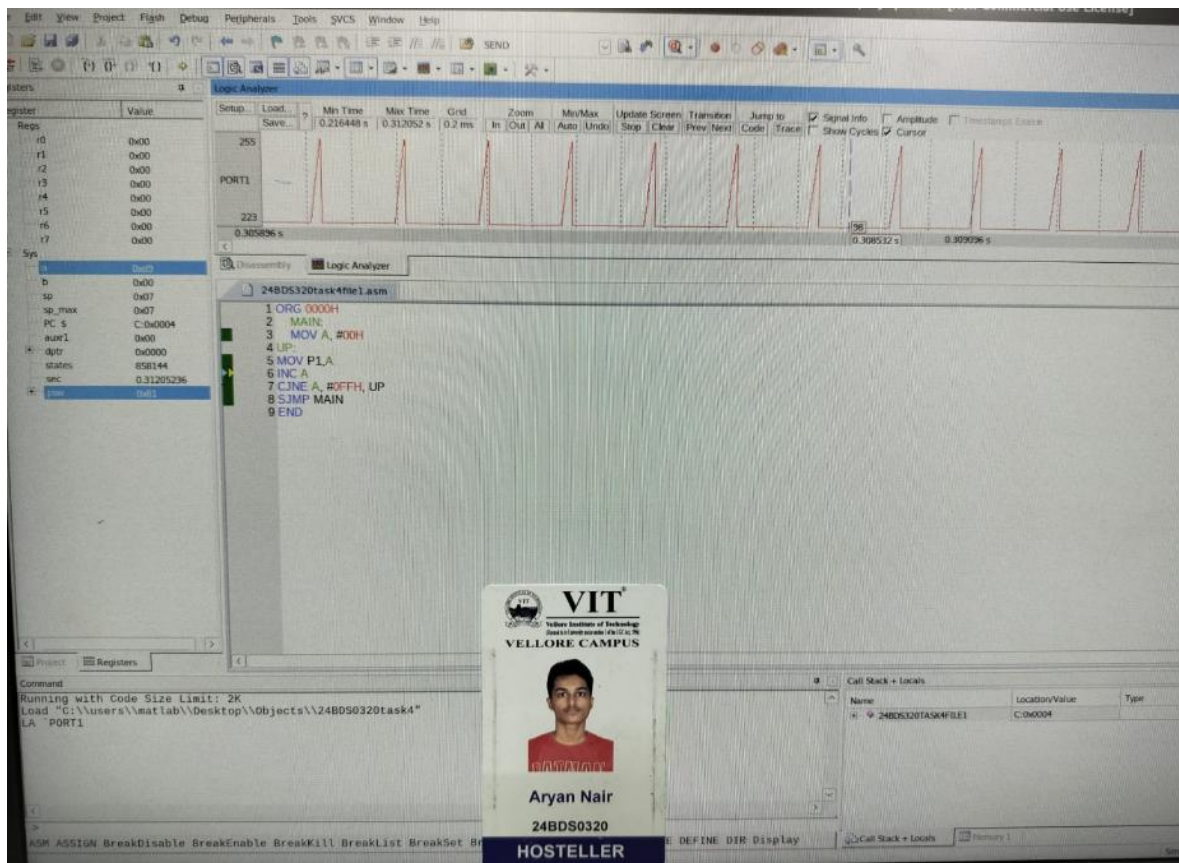
```
ORG 0000H
CLR P2.0
MOV R0, #00H
MAIN:
JB P1.0, WAIT1
INC R0
CJNE R0, #10, WAIT2
SETB P2.0
WAIT2:
JNB P1.0, WAIT2
WAIT1:
SJMP MAIN
END
```

OUTPUT:

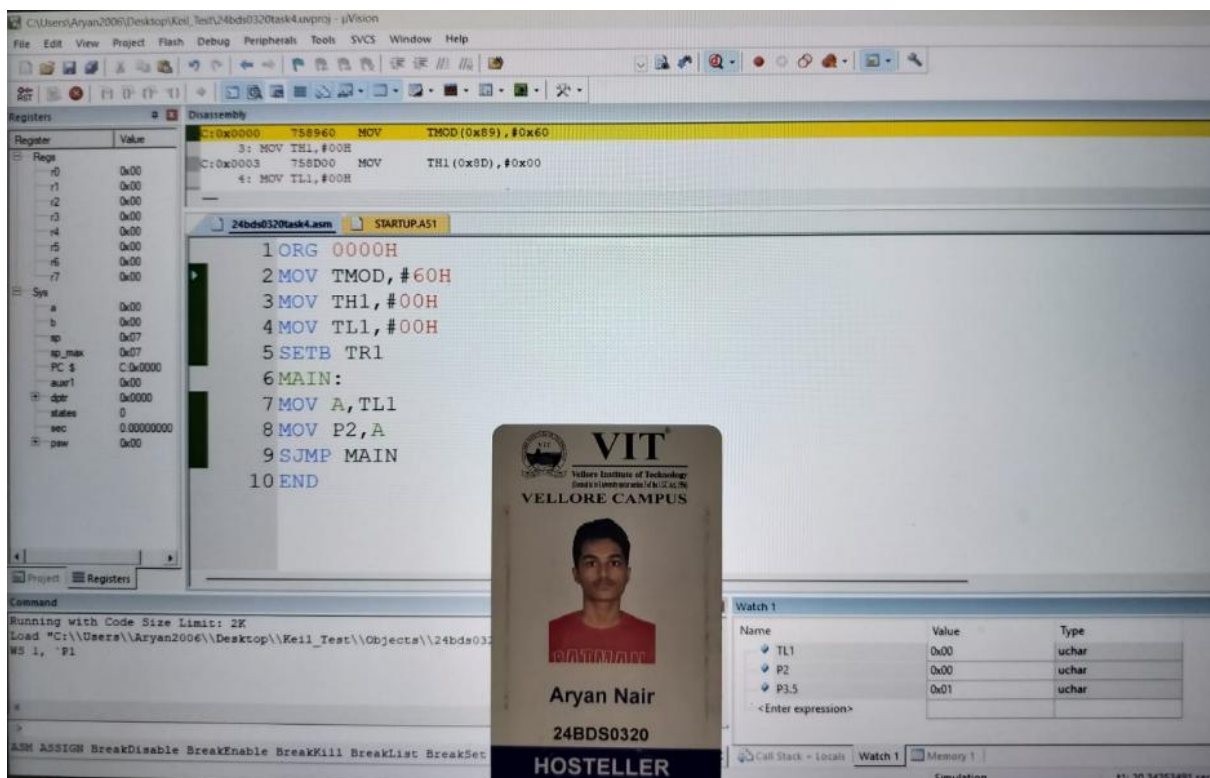
1)

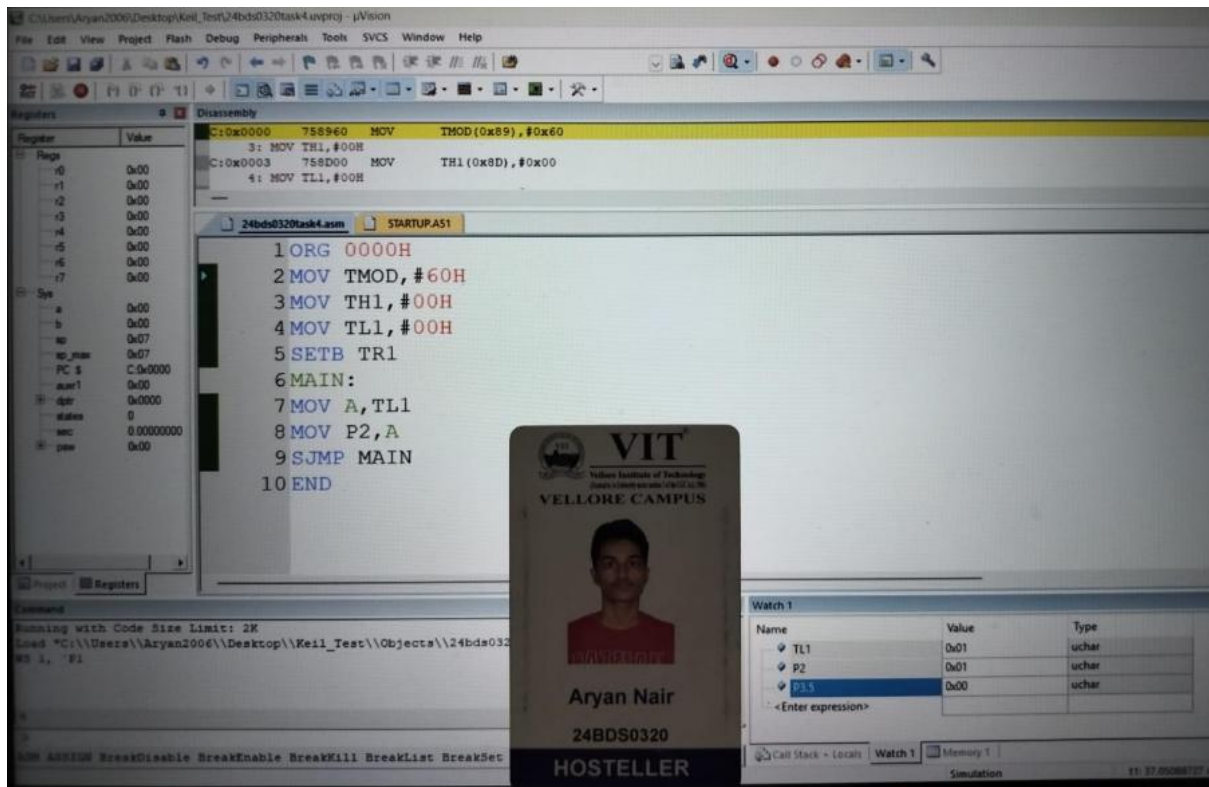


2)

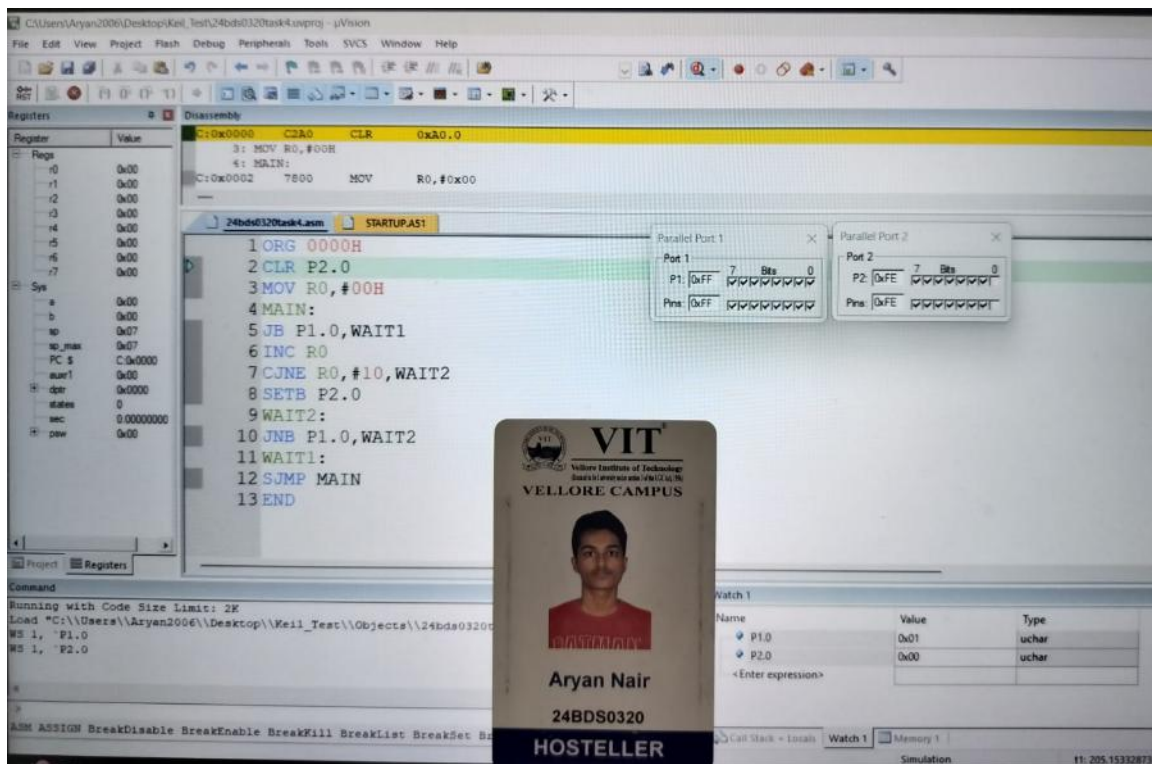


3)

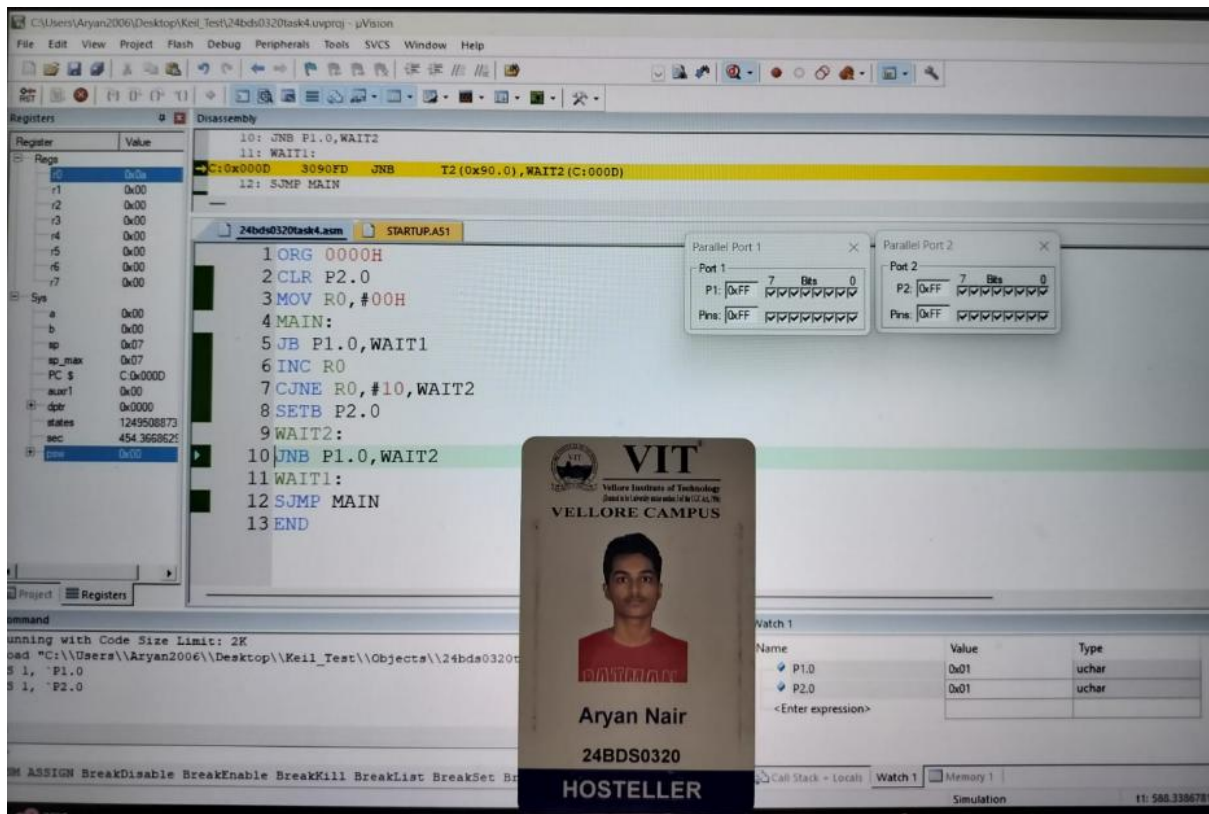




4)



Before starting toggle of P1.0



After 10 toggles of P1.0, we obtain P2.0=1 (10 people)

RESULT:

The given programs were written, compiled, and successfully executed using Keil μVision. The program produced the expected output, and the results were verified using the simulator by observing flags and register values.